

PAMUDI MENDIS

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SUMMARY

Recent Mechanical Engineering graduate from the University of Moratuwa with strong interests in metal additive manufacturing and advanced manufacturing technologies. Experienced in CAD modelling, finite element analysis, computational simulation, equipment design, and manufacturing process improvement through research, industrial training, and multidisciplinary projects. Proficient in SolidWorks, ANSYS, and AutoCAD, with research interests in process monitoring and control, process optimization, materials characterization, and sustainable manufacturing. Seeking PhD opportunities to investigate relationships among process parameters, material behaviour, and component performance, while contributing to intelligent, sustainable, high-performance manufacturing systems.

EDUCATION

University of Moratuwa

Mar 2022 – June 2026

B.Sc (Hons), Mechanical Engineering (CGPA 3.56/4.00)

IJSE (Institute of Software Engineering)

2020 – 2021

Comprehensive Master Java Developer

Southlands College, Galle

2012 – 2020

· G.C.E. A/L 2020: 3As – Physical Science Stream (z-score: 2.6142, Island Rank: 90)

· G.C.E. O/L 2017: 9As (Gold Medal for the Best Academic Achiever - O/L)

WORK EXPERIENCE

Visiting Instructor

Feb 2026 – Present

Department of Mechanical Engineering, University of Moratuwa

- Conduct practical sessions for undergraduate students in the following modules:

– Control Systems	Jul 2026 – Present
– Industrial Fluid Flow Systems	Jul 2026 – Present
– Mechanics of Materials II	Jul 2026 – Present
– Computer-Aided Design and Manufacture	Jul 2026 – Present
– Fundamentals of Engineering Thermodynamics	Feb 2026 – May 2026

Trainee Mechanical Engineer

Dec 2024 – June 2025

Hyundai Assembly Plant - Abans Auto (Pvt) Ltd

- Planned and coordinated assembly plant layout improvements to enhance workflow efficiency and optimize material movement.
- Analyzed manufacturing processes and redesigned the vehicle shower test system to improve process efficiency and reliability.
- Developed a detachable wheel mounting fixture that reduced forklift dependency and improved material handling efficiency.

PROJECTS

Design and Development of a Metal Additive Manufacturing System for Complex Component Fabrication

July 2025 – June 2026

Designed and developed a low-cost Wire Arc Additive Manufacturing system for fabricating complex metal components using MAG welding. Improved motion accuracy and process stability through control-system modifications, arc-length control, substrate cooling, parameter optimization, and thermo-mechanical simulation using ANSYS and MATLAB/Simulink. Validated the system experimentally through multi-layer deposition, dimensional evaluation, microscopic analysis, hardness testing, and fabrication of complex geometries.

Static Structural and Buckling Analysis of an Offshore Spreader Bar

May 2026 – June 2026

Developed a three-dimensional spreader bar model in SOLIDWORKS and performed static structural and linear eigenvalue buckling analyses in ANSYS. Evaluated deformation, von Mises stress distribution, mesh independence, reaction-force equilibrium, and critical buckling behaviour, and validated the results against published research.

Production line optimization at Abans Electricals PLC

March – May 2026

Optimized the washing machine production line at Abans Electricals PLC by identifying inefficiencies and proposing improvements in material handling, batch sizing, and workflow.

Design of an Ambulance Craft for Operations in the Northern Islands, Sri Lanka	<i>Sept – Dec 2025</i>
Designed a concept ambulance craft, including hull modelling, stability analysis, general arrangement, and propulsion selection, while ensuring compliance with marine standards.	
Development of a Below-Knee Prosthetic Leg Testing Machine	<i>Sept – Dec 2025</i>
Designed a low-cost, hand-operated mechanism to simulate gait cycles for static and fatigue testing of below-knee prosthetic limbs.	
Design of a Gearbox for a Domestic Coffee Grinder	<i>July – Oct 2024</i>
Designed a gearbox to enable multiple coffee grind sizes, optimizing the grinding process. Developed detailed 3D models and assemblies of the gearbox considering functionality and manufacturability.	
Optimization of the Production Line of Wewalwala Mills (Pvt) Ltd	<i>July – Oct 2024</i>
Analyzed existing production line conditions and identified inefficiencies. Proposed simulation-based improvements to enhance productivity and overall production line efficiency.	
Reverse Engineering of a Cloth Iron	<i>Mar – July 2024</i>
Identified materials and reverse-engineered all components of a domestic cloth iron. Developed 3D models using SolidWorks and performed FEA to recommend design improvements.	
Smart Waste Management System	<i>Mar – July 2023</i>
Developed a smart waste management system using image classification on a Raspberry Pi for waste sorting. Integrated Arduino-based compression and sealing mechanisms, real-time monitoring, and waste generation pattern analysis.	

PUBLICATIONS

Design and Development of a Metal Additive Manufacturing System Using Metal Inert Gas (MIG) Technology (Extended Abstract)

- Presented at the Engineering Research Unit Symposium 2025. Published in the conference proceedings.
- Available at: University of Moratuwa Digital Repository.

Towards Sustainable Metal Additive Manufacturing Using a Low-Cost MAG-Based WAAM System (Full Paper)

- Accepted for presentation at the 22nd Global Conference on Sustainable Manufacturing, September 2026.

Design and Modelling of a Closed-Loop Liquid Cooling System for Thermal Management of Wire Arc Additive Manufacturing (WAAM) Substrates Using MATLAB Simulink (Extended Abstract)

- Accepted for presentation at the 19th UK Heat Transfer Conference, September 2026.

HONORS & AWARDS

• Dean's List – Semester 7 (Semester GPA: 4.00) , University of Moratuwa	<i>Dec 2025</i>
• Dean's List – Semester 8 (Semester GPA: 4.00) , University of Moratuwa	<i>Dec 2025</i>
• Finalist – Chemistry Olympiad Sri Lanka 2020 , Institute of Chemistry Ceylon	<i>Feb 2020</i>
• Gold Medal for the Best Academic Achiever – G.C.E. O/L Examination 2017 , Southlands College	

LICENSES & CERTIFICATIONS

• Sustainability for Design, Construction, and Manufacturing	<i>Issued Jul 2025</i>
• Additive Manufacturing: Metal 3D Printing	<i>Issued Jul 2025</i>
• Simulink Onramp	<i>Issued Jul 2024</i>
• MATLAB Essential Training	<i>Issued Jul 2024</i>
• MATLAB Onramp	<i>Issued Jun 2024</i>

VOLUNTEERING EXPERIENCE

University of Moratuwa Mechanical Engineering Society - UMMES *Aug 2025 – Present* *Secretary*

- Contribute to UMMES initiatives through project coordination, people management, technical content development, social media management and event organization to enhance student engagement and professional development.

ECHELON 25 *June – July 2025* *Editorial Committee Lead*

- Organizing Committee Vice President - Marketing | Global Classroom 1.4 & Rooted 2.2
- Organizing Committee Member - Marketing | LEAD CS 10.0
- Organizing Committee Member - Partnership Development & Public Relations | HOPE 2023
- Organizing Committee Member - Events & Delegates | Global Village 2023
- Data Management Specialist - Finance & Legal Team

SKILLS

Programming Languages: Python, Java, C++, MATLAB

Software & Tools: SolidWorks, Solid Edge, ANSYS, AutoCAD, Siemens NX, MATLAB, Simulink

Languages: Sinhala, English

Soft Skills: Communication, People Management, Critical Thinking, Time Management, Fast Learner, Team Work

REFERENCES

Dr. Thamasha Samarasinghe

Senior Lecturer

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Dr. Eranga de Silva

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